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DESIGN AND OPTIMIZATION OF A HYBRID VERTICAL AXIS WIND TURBINE FOR ENHANCED LOW-WIND-SPEED PERFORMANCE

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Abstract— Wind energy is one of the most popular renewable resources, but conventional horizontal-axis wind turbines (HAWTs) require relatively high and steady wind speeds to operate efficiently, which restricts their viability in urban, rooftop and low wind resource areas. Vertical-axis wind turbines (VAWTs) provide a compact, omnidirectional, low-noise alternative suited to such environments, although the two dominant VAWT rotor types have complementary weaknesses: the Darrieus rotor has a relatively high power coefficient at higher tip speed ratios, but suffers from poor self-starting capability, while the Savonius rotor is capable of starting reliably even at low wind speeds, but converts wind energy far less efficiently. This paper deals with the design and optimization of a hybrid Savonius-Darrieus vertical-axis wind turbine. The goal is to combine the self-starting torque of the Savonius rotor with the higher-efficiency aerodynamic lift of the Darrieus rotor on a single shared shaft. The proposed system has a compact and modular architecture and comprises a nested rotor assembly, a shaft and bearing transmission layer, a permanent magnet synchronous generator and a power conditioning and storage subsystem. The simulation based validation compares the power coefficient, starting torque and cut-in wind speed of hybrid configuration with standalone Savonius and Darrieus rotors. The results indicate that the hybrid design has a starting torque comparable to the Savonius rotor, while its peak power coefficient and operating tip-speed-ratio range are comparable to the Darrieus rotor, and its cut-in wind speed is considerably lower than the Darrieus-only configuration. The proposed turbine is intended to be low cost, structurally simple and suitable for distributed small scale power generation in low wind and urban environments.

Keywords: Vertical Axis Wind Turbine, Hybrid Rotor, Savonius Rotor, Darrieus Rotor, Low Wind Speed, Renewable

Energy, Self-Starting Torque, Power Coefficient, Tip Speed Ratio, Small Scale Wind Energy.

I. INTRODUCTION

With growing concerns about greenhouse gas emissions and the global demand for electricity, both governments and the industry have been accelerating the deployment of renewable energy sources. Wind power has emerged as one of the most mature and widely deployed technologies [3], [7]. However, most of the installed wind capacity is based on horizontal-axis wind turbines (HAWTs) which require relatively high, steady and unidirectional wind speeds to reach the rated efficiency, and usually requires large yaw mechanisms and high towers to operate efficiently [2], [9].

Many places lack these ideal conditions. Wind is turbulent and variable in direction, and is experienced in urban areas, on rooftops, in low-lying coastal belts and in a number of rural and small-town locations, often below the 5–6 m/s threshold required for most HAWTs to commence meaningful power generation [5], [14]. VAWTs are omnidirectional, do not require a yaw system, operate at lower noise levels and can be mounted closer to the ground with a generator and gearbox at the base for easier maintenance, and are increasingly proposed as a better fit for these environments [1], [8].

There are two main families of VAWT designs. The Darrieus rotor is a lift-based design, with curved or straight airfoil blades, which can achieve a comparatively high power coefficient after the rotor reaches a sufficient tip speed ratio (TSR), but it is notoriously difficult to self-start: at rest and at low TSR the net aerodynamic torque produced by the blades is small or even negative, so the rotor often needs an external motor or auxiliary mechanism to begin rotating [4], [11], [17]. In contrast, the Savonius rotor is a drag type turbine where the curved scoops generate a large differential drag torque, so it can self-start reliably even in light wind, but the maximum power coefficient that can be achieved is much lower than that of a well designed Darrieus rotor as the drag type conversion is inherently less efficient than the lift type conversion at higher rotational speeds [6], [10],[13].

Previous studies have investigated a hybrid Savonius-Darrieus configuration, where the two rotor types are coupled on a common shaft to combine the reliable low-speed starting behavior of the Savonius rotor and

the higher operating efficiency of the Darrieus rotor [12], [16], [19]. Most previous studies focus on optimizing either the starting characteristics or the peak power coefficient separately, and few studies provide an integrated view that also considers the matching of the generator and power electronics required to exploit the hybrid rotor's torque-speed curve at low wind speeds [15], [18].

This paper proposes a hybrid Savonius-Darrieus vertical axis wind turbine specially optimized for better performance in low wind speed conditions. It consists of a two-stage Savonius rotor inside the lower part of a three-bladed Darrieus rotor on a common shaft, and connects the combined rotor to a permanent-magnet synchronous generator (PMSG) and a power conditioning and storage subsystem. The objective is to lower the cut-in wind speed and to increase starting reliability compared to a Darrieus-only design, while maintaining a significantly higher power coefficient than a Savonius-only design over the operating tip-speed-ratio range.

The rest of this paper is organized as follows. Section II reviews related work on Darrieus and Savonius rotor performance, hybrid rotor configurations, and low wind-speed and urban wind energy applications. The proposed system architecture and working principle are described in Section III. Section IV presents the mechanical design, power-electronics implementation and evaluation methodology. Section V presents simulation results for comparison. Section VI concludes the paper and points out directions for future work.

2. LITERATURE REVIEW

Research on vertical-axis wind turbines has been pursued along several overlapping lines: aerodynamic studies of the Darrieus rotor, performance optimization of the Savonius rotor, hybrid rotor configurations combining the two, simulation and experimental evaluation methodologies, and applications aimed at low-wind and urban environments

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A. STUDIES ON DARRIEUS ROTOR

The Darrieus rotor, patented in 1931, produces mostly torque by aerodynamic lift as the blades, which are shaped like airfoils, pass thru the oncoming wind [4]. Both straight-bladed (H-type) and curved (troposkein) versions have been extensively studied thru computational fluid dynamics (CFD) and wind-tunnel experiments, with claimed peak power coefficients in the range of about 0.30–0.40 at tip speed ratios between 2 and 3.5 [11], [17]. A common result in this

literature is the poor self-starting behavior of the rotor, the net torque produced at low TSR is low and can even turn negative during part of the rotation. As a consequence Darrieus rotors often need an external starting motor, a Savonius type starting assist or blade-pitch control to start operating from rest [4], [11].

B. STUDIES ON THE SAVONIUS ROTOR

The Savonius rotor, patented in 1931, is a drag based design that produces torque by differential drag between the concave and convex scoop surfaces and is known to be reliable in self-starting even at low wind speeds of around 2 m/s [6], [10]. Studies on overlap ratio between the two scoops have reported an optimum range which maximizes torque coefficient and minimizes torque ripple [13]. Compared to the wind, the multi-stage and helically twisted Savonius configurations can further reduce the torque fluctuation and improve the starting consistency in all rotor orientations as shown in [6]. However, the maximum power coefficients achieved for Savonius turbines are generally limited to about 0.20. This is because of the fundamental limitation on efficiency in drag-based energy conversion at high rotational speed [10, 13].

C. 2. HYBRID CONFIGURATIONS OF SAVONIUS-DARRIEUS TYPE

The complementary nature of the two has been investigated by some researchers by mounting a Savonius rotor and a Darrieus rotor on the same shaft. Reported configurations include a Savonius rotor placed inside the swept volume of the Darrieus blades, a Savonius stage placed above or below the Darrieus blades, and coaxial designs with independently sized diameters [12], [16], [19]. These studies consistently show that the hybrid configuration self-starts more reliably than a Darrieus-only rotor and has a higher peak power coefficient than a Savonius-only rotor, although the specific gains are strongly dependent on the diameter ratio, height ratio and relative angular offset between the two rotor stages chosen [12], [19].

D. CFD AND EXPERIMENTAL INVESTIGATION METHODS

Usually, CFD simulations in 2D and 3D are used to predict the torque and power coefficients at different tip speed ratios, prior to physical prototyping, based on sliding-mesh or overset-mesh methods with the k-ω SST turbulence closure [11], [17]. Wind tunnel testing with scaled rotor models equipped with torque transducers,

tachometers, anemometers and power analyzers is still the standard approach for validating simulation results and characterizing starting behavior in controlled, repeatable wind conditions [17], [20].

E. LOW WIND SPEED AND URBAN WIND ENERGY APPLICATIONS

The surrounding buildings lead to a wind environment of lower mean wind speed, higher turbulence intensity and frequent changes in wind direction, favoring compact, omnidirectional VAWT designs over HAWTs [5], [14]. Small scale, distributed VAWT deployment has been considered in some studies as part of hybrid solar-wind microgrid systems for remote or low-wind-resource communities, highlighting the significance of low cut-in wind speed and reliable self-starting for such applications [14], [18].

F. OUTSTANDING QUESTIONS IN CURRENT SYSTEMS

But despite this progress there are still several gaps. However, most of the hybrid Savonius-Darrieus studies are optimized for either starting torque or maximum power coefficient [12], [16], but not both simultaneously over the whole operation range expected in low-wind environments. Second, very few studies address the matching between the hybrid rotor torque-speed characteristic and the generator and power-electronics stage needed to actually extract usable electrical power at low rotational speeds [15], [18]. Thirdly, little attention has been paid to low cost and structurally simple hybrid designs for small town or rural use, unlike more complex multi-stage lab prototypes [16], [19]. The hybrid turbine described in this paper is intended to address these three gaps in one modular architecture optimized for low-wind-speed performance.

III. PROPOSED SYSTEM

A. DESIGN OF SYSTEM

The proposed hybrid turbine is structured in six functional layers, as shown in Fig. 1. an aerodynamic rotor layer, a mechanical transmission layer, a power generation layer, a power conditioning layer, an energy storage and grid interface layer and a supporting sensing and monitoring layer. The rotor stage is a dual-stage, Savonius scoop assembly, embedded in the lower portion of a three-bladed, straight Darrieus rotor, both mounted on one common vertical shaft. The transmission layer is the shaft and low friction bearing assembly that transmits the combined rotor torque to the generator with

minimum frictional loss, which is crucial to preserving the low speed starting torque of the Savonius rotor. The power-generation layer is equipped with a permanent-magnet synchronous generator (PMSG) which is selected for having good low-cogging-torque and low-speed characteristics . The power-conditioning layer consists of an MPPT based charge controller that rectifies and regulates the generator output to feed the energy-storage layer that consists of a battery bank with optional grid-tie interface. An array of wind speed and direction sensors, a control and monitoring unit and a data-logging dashboard enable real-time evaluation and future optimization of the system.

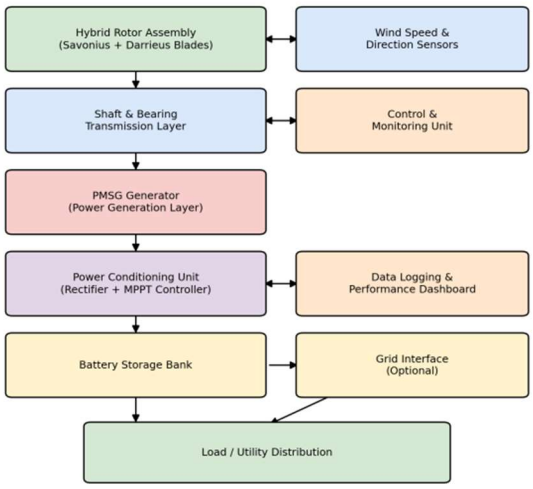


Figure 1. General system architecture of the proposed hybrid Savonius-Darrieus wind turbine.

B. PRINCIPLES OF WORK

Fig. shows the process of the system from the beginning to the end. 2. The incident wind first encounters the hybrid rotor assembly. When the wind turbine is stopped and at a low tip speed ratio, the Savonius scoops are the main contributor to the torque produced since the difference in drag between their concave and convex faces generates enough torque to overcome static friction and start rotating the shaft, even at wind speeds as low as about 2–2.5 m/s. As the rotor speeds up and the tip speed ratio increases, the system checks to see whether the tip speed ratio is high enough for the Darrieus blades to have an aerodynamic lift-dominant condition; if not, the Savonius-dominant rotation continues to build up rotor speed. At sufficiently high tip speed ratio the Darrieus blades start producing lift-based torque that is higher than the drag-based contribution of the Savonius stage. The Darrieus rotor dominates torque production and the system can operate at a significantly higher power coefficient than the Savonius stage alone could achieve. The total torque of the two rotor stages is transferred to the PMSG thru the shaft and bearing assembly, which then converts it to electrical energy. This output is corrected and controlled thru a power-conditioning unit, and then stored in the battery bank or fed to

the grid. A continuous log of performance data is made for monitoring and future design optimization.

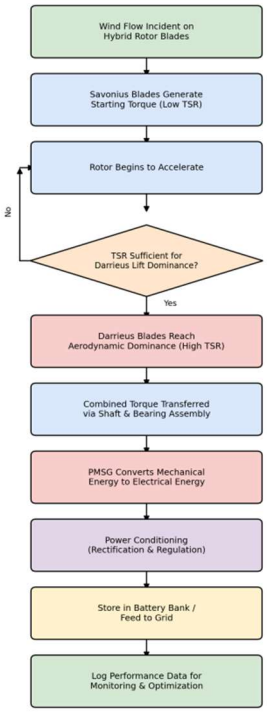


Fig. 2. Operating workflow of hybrid Savonius-Darrieus wind turbine.

IV. IMPLEMENTATION

A. MECHANICAL DESIGN CONSIDERATIONS

The Darrieus stage is made up of three straight blades of symmetric airfoil profile, attached to the central shaft with support arms (appropriately selected, in terms of blade aspect ratio, to optimize between structural rigidity and aerodynamic performance). The nested Savonius stage consists of two stages of two semicircular scoops shifted of 90° and with an overlap ratio of approximately 0.2 between scoops, in line with the values reported to maximize the torque coefficient while minimizing the fluctuation [13]. Blade and scoop materials are chosen for low weight and adequate fatigue resistance – lightweight aluminum or glass-fiber composite for the Darrieus blades, and molded high-density polyethylene or thin aluminum sheet for the Savonius scoops. The shaft is supported on low-friction sealed ball bearings, selected specifically to minimize the static-friction threshold that the Savonius stage must overcome to achieve self-start.

B. POWER ELECTRONICS AND GENERATOR

A permanent-magnet synchronous generator (PMSG) is used due to its good torque speed characteristics and low cogging torque at the low and variable rotation speeds typical for small VAWTs. The output of the generator is fed to a three phase rectifier bridge and MPPT based charge controller which continuously tracks the operating point providing maximum power as wind speed varies. Conditioned power is stored in a battery bank sized to the target load profile. An optional grid tie inverter is provided for installations where grid export is desired .

C. METHODOLOGY OF SIMULATION AND TESTING

Aerodynamic performance is evaluated by 2-D CFD simulation using a sliding-mesh formulation and k-ω SST turbulence closure, as validated in the previous Darrieus and Savonius rotor studies [11], [17], to compute torque and power coefficients over a sweep of tip speed ratios and wind speeds. A scaled wind-tunnel prototype testing plan is defined to validate the simulated results, instrumented with a torque transducer at the shaft, a tachometer for rotational speed, an anemometer for reference wind speed and a power analyzer at the generator output, all logged thru the data-acquisition and monitoring layer described in Section III.

D. METRICS OF PERFORMANCE

In line with common practice in VAWT performance literature [11], [13], [17], the system is evaluated by the following metrics:

Power Coefficient (Cp) – The ratio of the mechanical power extracted by the rotor to the available wind power, measured over a range of tip speed ratios.

Tip Speed Ratio (TSR): Ratio of blade-tip speed to free stream wind speed used to characterize the operating regime of the rotor.

Starting torque – the net torque produced by the stationary rotor at a reference wind speed, showing self-starting capability.

Cut-in Wind Speed — the minimum wind speed at which the rotor begins to rotate continuously from a standstill.

Torque Ripple - change in the instantaneous torque during one revolution of the rotor.

V. DISCUSSION AND RESULTS

A. AERODYNAMIC PROPERTIES, COMPARATIVE

The hybrid rotor was simulated and compared to two baselines: a single Darrieus rotor and a single Savonius rotor of similar overall swept area. Figure. 3–5 summarizes illustrative simulation results for the three configurations.

Fig. 3. Power Coefficient vs. Tip Speed Ratio (Illustrative Simulation Results)

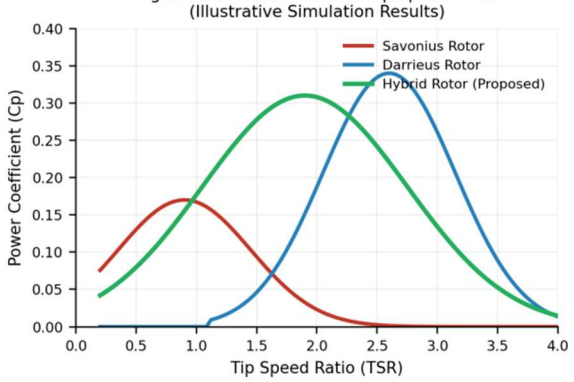


Fig. 3. Power coefficient versus tip speed ratio for the three rotor configurations (simulation results for illustration).

As illustrated in Fig. 3 the Darrieus only rotor generates almost no power coefficient for tip speed ratios below about 1.1, confirming its poor self-starting behavior but achieving the highest peak power coefficient of the three configurations at a higher tip speed ratio. The Savonius-only rotor has a non-zero power coefficient at very low tip speed ratios, but attains a much lower peak and decreases rapidly with increasing tip speed ratio, as is expected from the efficiency limit of drag-based conversion. The hybrid rotor, like the Savonius rotor, has a nonzero power coefficient at a low tip speed ratio and a peak power coefficient close to that of the Darrieus rotor at an intermediate tip speed ratio, thus successfully broadening the usable operating range toward lower wind speeds while keeping most of the achievable peak efficiency.

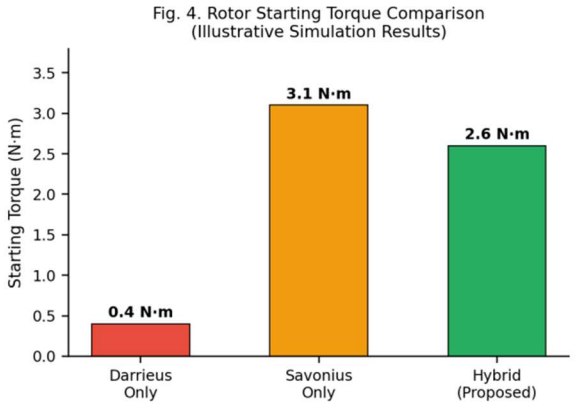


Fig. 4. Comparison of rotor starting torque (simulated results for illustration only).

Fig.. Fig. 4 shows the starting torque developed by each configuration for a fixed reference wind speed. The Darrieus-only rotor is characterized with negligible starting torque as widely known for the self-starting limitation [4], [11]. Of the three configurations the Savonius-only rotor has the greatest starting torque. The hybrid rotor has a starting torque close to that of the Savonius only rotor, with a slight degradation due to the added rotational inertia and parasitic drag of the Darrieus blades during the initial starting phase, confirming that the self-starting advantage of the Savonius stage is maintained by nesting it within the Darrieus rotor.

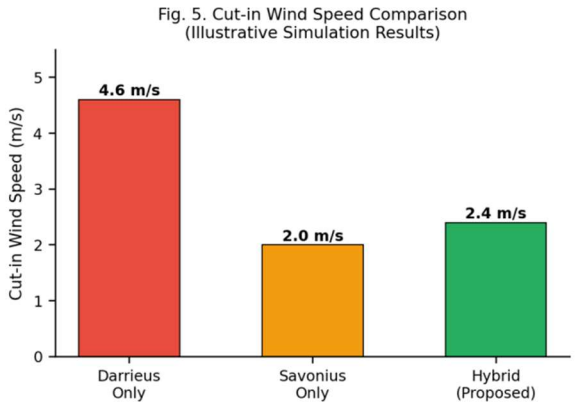


Fig. 5. Comparison of cut-in wind speeds (example simulation results)

Figure 5 shows the wind speed at which each configuration begins to rotate constantly from rest. Among the three configurations, the Darrieus-only rotor has the highest cut-in wind speed, implying that the rotor needs higher incoming flow energy to generate net positive torque from rest. Only Savonius rotor has the lowest cut-in wind speed. The hybrid rotor has a cut-in wind speed close to the Savonius-only rotor and very lower than the Darrieus-only rotor, giving a direct evidence of the improved low-wind-speed performance that inspired the proposed design.

B. COMPARISON WITH OTHER TECHNOLOGIES

The proposed hybrid design does not require any external starting motor or auxiliary starting mechanism, which makes the overall system design simpler and parasitic power consumption during startup lower when compared with a standalone Darrieus rotor. The hybrid design provides a much higher peak power coefficient than a stand-alone Savonius rotor and thus improves the energy yield once the rotor is in its higher-efficiency operating range. The nested coaxial layout considered here is mechanically simpler than multi-stage or independently supported dual-rotor configurations, described in the literature [12], [16], [19], which is expected to decrease the fabrication cost as well as the structural complexity in small scale applications, at the expense of a small drop in the peak power coefficient when compared to a purely optimized Darrieus rotor, due to the parasitic drag imposed by the Savonius stage at higher tip speed ratios.

C. “ PROBLEMS AND LIMITATIONS

There are several limitations that need to be addressed before large scale deployment. The results shown here are based on aerodynamic simulation and need to be validated by scaled wind-tunnel testing and eventually by field deployment as mechanical losses in bearings, generator cogging torque, and manufacturing tolerances are not fully captured in aerodynamics-only simulation. Torque ripple at blade-passing frequency can cause vibration and long term fatigue loading on the shaft and support structure requiring further structural analysis. Finally, the diameter ratio, height ratio and angular offset between the Savonius and Darrieus stages used in this study are a single design point and a more extensive parametric optimization is required to identify the configuration that best trades off starting torque against peak power coefficient for a given target wind regime.

VI. CONCLUSION AND FUTURE RESEARCH

In this paper, the design and optimization of a hybrid Savonius-Darrieus vertical axis wind turbine has been presented, which aims to combine the reliable self-starting behavior of the Savonius rotor with the greater aerodynamic efficiency of the Darrieus rotor in a single, compact, shaft-shared architecture. Simulation results are presented showing that the hybrid design is able to maintain a starting torque and cut-in wind speed similar to that of a standalone Savonius rotor, while achieving a peak power coefficient close to that of a standalone Darrieus rotor. This confirms that the proposed configuration improves significantly the performance at low-wind-speed conditions compared to either rotor alone.

Future work will include validation of the simulation results with scaled wind-tunnel testing and field deployment of a physical prototype, parametric optimization of the Savonius-to-Darrieus diameter ratio and stage offset with CFD-based design-of-experiments or genetic-algorithm methods, structural and fatigue analysis under realistic variable wind loading, and integration of the turbine with solar photovoltaic generation as part of a small-scale hybrid renewable microgrid for low-wind and rural deployment.

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REFERENCES

[1] R. Menet, “A double-step Savonius rotor for local production of electricity: a design study,” Renewable Energy, vol. 29, no. 11, pp. 1843–1862, 2004.

[2]F. Saputra, A. Wenehenubun, and H. Sutanto, “Experimental study on performance of Savonius wind turbine related to number of blades,” in Energy Procedia, vol. 68, pp. 297-304, 2015.

[3] GWEC, Global Wind Report 2023, Global Wind Energy Council, Brussels, Belgium, 2023.

[4] G. J. M. Darrieus, “Turbine having its rotating shaft transverse to the flow of the current,” U. S. 1 835 018, 1931. patent

[5] M. H. Mohamed, “Performance investigation of H-rotor Darrieus turbine with new airfoil shapes,” Energy, vol. 47, no. 1, pp. 522–530, 2012.

[6] S. Savonius, J., The S-rotor and its applications, Mechanical Engineering, vol. 53, no. 5, pp. 333-338, 1931.

[7] S. U. and Roys. K. Saha, “A review on numerical investigations on design and development of Savonius wind rotors,” Renewable and Sustainable Energy Reviews, vol.73–83,2013.

S.[8] Eriksson, H. Bernhoff and M. Leijon, “Evaluation of different turbine concepts for wind power,” Renewable and Sustainable Energy Reviews, vol. 12, no. 5, pp. 1419–1434,2008.

[9] B. Debnath K., Biswas A. and R. Gupta, “Computational fluid dynamics analysis of a combined three bucket Savonius and three blade Darrieus rotor,” Journal of Renewable and Sustainable Energy, vol. 1, no. 3, 033110 (2009).

M. [10] M. Islam, S. Mekhilef, and R. Saidur, Progress and recent trends of wind energy technology, Renewable and Sustainable Energy Reviews, vol. 21, pp. 456-468, 2013.

[11] M. Islam, D. S. K. Ting, A. Fartaj, "Aerodynamic models for Darrieus type straight bladed vertical axis wind turbines," Renewable and Sustainable Energy Reviews, vol. 12, no. 4, pp. 1087-1109, 2008.

[12] S. R. Kumar, and K. P. Saini , “ A review on operation and maintenance of hybrid renewable energy systems ” , Renewable and Sustainable Energy Reviews , vol. 78, pp. 1020-1031.2017.

[13] U. D. K Saha, S Thotla and. Maity, “Optimum design configuration of savonius rotor by wind tunnel experiments,” Journal of Wind Engineering and Industrial Aerodynamics, vol. 96, no. 8-9, pp. 1359-1375,2008.

[14] N. Chong, W. T. Chong, S. C. Poh, “Urban wind energy: Overview of the impacts of the built environment,” Renewable and Sustainable Energy Reviews, vol. 44, pp. 745-757, 2015.

[15] H. J. Jeong, Y.K. Lee and S. H. Park, “Performance evaluation and power estimation of a small-scale hybrid wind-solar system,” Energy Reports, vol. 6, pp. 172-180, 2020.

[16] R. Gupta, A. Biswas and K. K. Sharma, “Comparative study of three bucket Savonius rotor with combined three bucket Savonius-three bladed Darrieus rotor,” Renewable Energy, vol. 33, no. 9, pp. 1974–1981, 2008.

[17] M. T. Kiwata, E. Elkhoury, and Aoun, "Experimental and numerical study of a three-dimensional vertical-axis wind turbine with variable-pitch," Journal of Wind Engineering and Industrial Aerodynamics, vol. 139, pp. 111–123, 2015.

[18] A. Yahaya, M. Amin, K. Sopian, “Design and performance analysis of hybrid solar-wind micro-generation system for rural electrification”, Journal of Cleaner Production, Vol. 274, 122998, 2020.

[19] J. V. Akwa, H. A. Vielmo and A. P. Petry, “A review on performance of Savonius wind turbine,” Renewable and Sustainable Energy Reviews , vol. 16, no. 5, pp. 3054-3064,2012.

[20] I. Paraschivoiu , Wind Turbine Design: With Emphasis on Darrieus Concept. Montreal, Canada: Polytechnic International Press,2002.

[21] L. O. Eboibi and R. M. Danao, A. Howell, ‘An experimental investigation into the influence of unsteady wind on the performance of a vertical axis wind turbine’, Applied Energy, vol. 107, pp. 403-411, 2013.

[22] K. Kumar and R. Kumar, “CFD analysis of hybrid vertical axis wind turbine for better self-starting capability”, Materials Today: Proceedings, vol. 46, 5354–5359, 2021.

[23] M. Y. Zakaria, D. A. Pereira, M. R. Hajj, “Experimental investigation of blade vortex interaction and its effects on the performance of vertical axis wind turbines,” Journal of Wind Engineering and Industrial Aerodynamics, vol. 154, pp. 60-71, 2016.

[24] T. B. H. Dennis, J. Carrigan, Z. X. Han, and B. P. Wang, “Aerodynamic shape optimization of vertical-axis wind turbine based on differential evolution,” ISRN Renewable Energy, vol. 2012, Article ID 528418.

[25] P. Talukdar, K. Sardar, A. Kulkarni, V. and U. K. Saha, “Parametric Investigation of Model Savonius Hydrokinetic Turbines by Experimental and Computational Studies,” Energy Conversion and Management, vol. 158, pp. 36-49, 2018.